Deep Encoder With Kernel-Wise Taylor Series for Hyperspectral Image Classification

Changda Xing, Jianlong Zhao, Chaowei Duan, Zhisheng Wang, and Meiling Wang

***Abstract*— Deep learning is a popular and effective technique for hyperspectral image (HSI) classification. Current deep-learning-based methods have numerous free parameters to be trained. They may be unavailable once lacking training samples. In addition, these approaches only use features from the deepest layer and exclude shallow features, which is a kind of information loss. To remedy such deficiencies, in this work, we construct a novel deep encoder with kernel-wise Taylor series (EKTS) for HSI classification. More specifically, we introduce the Taylor series to approximate the role of deep networks for feature extraction. Because the original Taylor series is linear, the kernel theory is used to build EKTS to encode the HSI data and extract deep nonlinear features. Furthermore, an alternating iterative optimization strategy is developed to obtain the outputs of all the layers of the built deep encoder. Subsequently, we stack the outputs of all the layers to obtain the final features that integrate both shallow and deep features. Finally, support vector machine (SVM) is adopted to deal with the obtained final features so that label results can be predicted. Unlike current deep learning methods, our EKTS has no free parameters to be trained and combines the advantages of both shallow and deep features to predict labels. Sufficient experimental analysis has been performed to verify greater classification performance of our EKTS method compared with many state-of-the-art approaches.**

***Index Terms*— Deep encoder, hyperspectral image (HSI) classification, kernel, Taylor series.**

I. Introduction

As A rising technique, hyperspectral image (HSI) classification integrates both the artificial intelligence technology and remote sensing science to predict class labels of ground materials, which has achieved great success in many Earth observation tasks such as urban mapping, environmental management, and forest monitoring [1], [2], [3], [4]. How to effectively and efficiently extract valuable features from complex HSIs is much critical in such classification problem.

To acquire features for HSI classification, numerous approaches have been proposed in the past years. Among them, spectral feature-driven methods [5], [6], [7] are classical. For example, in [5], the presented method uses raw spectral features (RAWs) only as discriminative features, and support vector machine (SVM) is adopted for label prediction (RAW-SVM). However, these spectral-feature-driven methods for classification fail to obtain high performance, since rich spatial and spectral information are included in HSIs. To describe the spatial information, many spatial-feature-based approaches have been developed such as Gabor filter [8], wavelet transform [9], gray-level co-occurrence matrix [10], and mathematical morphology [11]. In particular, the morphological methods have provided a new avenue to achieve good classification results, which generally combine the opening and closing operations to describe the spatial structure. For instance, extended morphological attribute profile (EMAP) [12] has been proposed to automatically build the extended attribute profiles with the standard deviation attribute based on the statistics of the samples belonging to the classes of interest, which has been experimentally validated to succeed in HSI classification. However, the features produced by these methods are shallow, and they suffer from low robustness for complex conditions, with the consequence that an image may change a lot even within a short interval under the sharply varied imaging environment [13].

It is well-known that deep learning is an available tool to obtain high-level features of data in a deep manner [14], [15]. Unlike shallow features, high-level features are able to describe more complex structure information. Thence, deep learning is robustness and invariance toward local variations in the input data. In the HSI classification field, deep learning has also obtained competitive performance [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31]. The idea of deep learning was for the first time introduced for HSI classification in [16], where the stacked autoencoder (SAE) is used to represent high-level features of HSI data, and logistic regression is further used for label prediction. In [17], a framework named compact and discriminative SAE (CDSAE) was proposed, which can learn discriminative low-dimensional feature mappings and train an effective classifier progressively. Chen et al. [18] proposed to use a deep belief network (DBN) to extract

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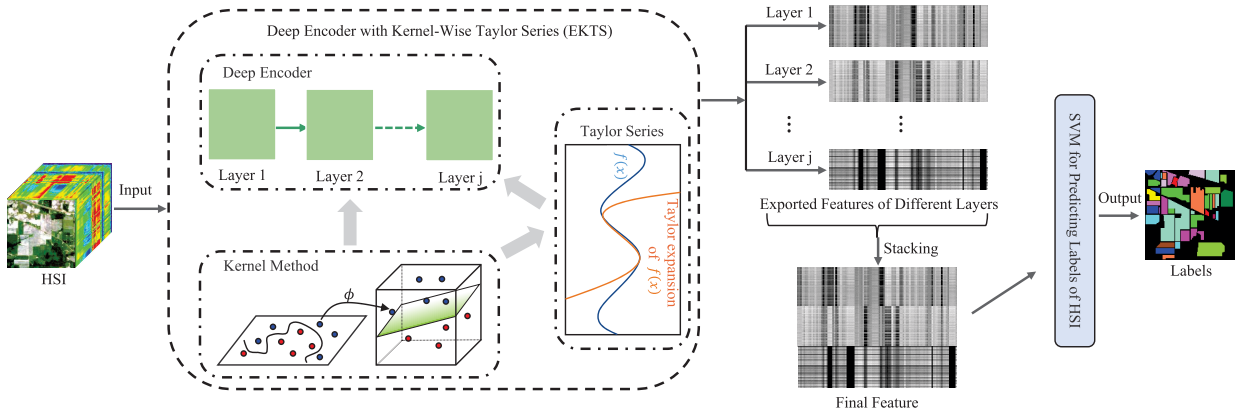


Fig. 1. Overall framework of the proposed deep encoder with EKTS for HSI classification.

deep features for classification, where single-layer restricted Boltzmann machine and multilayer DBN framework are integrated together. Note that [16], [17], and [18] are capable of capturing both deep spectral and spatial features, which achieve accurate classification accuracy. Similarly, convolutional neural networks (CNNs) are greatly able to naturally describe deep spectral and spatial features. For example, in [20], a CNN-based dimension reduction approach was proposed for simultaneously mining spectral and spatial features, which were further integrated together to predict labels. Cao et al. [21] presented an impressive framework combining CNN and active learning (AL) together (CNN-AL), which benefits from the high discriminant capability of deep learning and labeling efficiency of AL. In [26], a squeeze multibias network (SMBN) was proposed, where an efficient squeeze convolution module is introduced to efficiently improve the running speed of the network by reducing the number of parameters, while the high classification accuracy is still maintained. More advanced, Shen et al. [27] developed an efficient nonlocal module-based deep fully convolutional network (ENL-FCN) to incorporate the long-range contextual information for HSI classification. Xu et al. [28] proposed a robust self-ensembling network (RSEN) to address the classification task. Besides the above traditional deep learning approaches, there also are many hierarchical feature-extraction-based methods for HSI classification. For instance, in [29] and [30], the principal component analysis (PCA) was used to learn the convolution kernels from a series of patches, and they were further used for obtaining the features in a hierarchical way. In [31], the dropout technique was introduced into dictionary learning with k-means clustering to obtain deep features of the given HSI.

In spite of great performance achieved by these deep learning approaches, some deficiencies should be further considered. *First*, current deep networks contain numerous free parameters, which require to be estimated via adding sufficient training samples. When the number of samples is limited, these deep networks may become unsatisfactory, even unavailable. How to design a network, which excludes free parameters determined by training, is the *first motivation* of this work. *Second*, to the best of our knowledge, nearly all the current deep learning methods only use the deepest features (i.e., features from the deepest layer) for completing the classification task, so that shallow features are neglected. This neglect can be treated as a kind of information loss. It must be very valuable and meaningful for HSI classification that both deep and shallow features are integrated together. Therefore, we will find a way to simultaneously use deep and shallow features to classify HSIs, which is the *second motivation* of this work.

Motivated by the above-mentioned description, in this article, we construct a deep learning method, called deep encoder with kernel-wise Taylor series (EKTS), for the HSI classification task. The overall framework is shown in Fig. 1. To be specific, to avoid free parameters that need to be trained, we introduce the Taylor series to approximate the role of deep networks as a way to extract deep features. Since the direct use of the Taylor series can only extract linear features of the input HSI, the kernel theory is adapted to expand the above deep Taylor series for feature extraction, which aims to effectively describe deep nonlinear features of the data. Furthermore, an alternating iterative optimization strategy is designed to optimize the constructed deep encoder model with EKTS and obtain the outputs of all the layers. Considering the combination of both deep and shallow features, the outputs of all the layers are stacked to determine the final feature form. Finally, the SVM classifier is used to process the final features and predict the results. We assess the performance of the proposed EKTS method on three public HSI datasets, i.e., Salinas, Indian Pines, and Pavia Centre, with the results demonstrating its effectiveness and superiority in the task of HSI classification.

The main contributions of this article are fourfold. *First*, we introduce the Taylor series to approximate the role of deep networks for extracting deep features of HSIs, which excludes any free parameters that need to be trained. *Second*, we use the kernel theory to expand the deep Taylor series, aiming to obtain deep nonlinear features of HSIs. *Third*, we design an alternating iterative optimization strategy to get the outputs of all the layers of the built deep encoder based on EKTS. *Fourth*, we propose to integrate both shallow features and deep features together for label prediction.

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The remainder of this article is organized below. In Section II, we simply present preliminaries on the Taylor series. The developed EKTS approach is introduced in Section III. The experimental results on three datasets and related discussion are presented in Section IV. The conclusions of this article are given in Section V.

II. Preliminaries on Taylor Series

As we all know, the Taylor series theory [32], [33], [34] provides an approximation of an i times differentiable function around a defined point by an i-order Taylor polynomial. Actually, the Taylor theory can be used to analyze important information contained in the input data. It often decomposes the input into several parts, and each part can be regarded as a feature map. Such feature map reflects the characteristics of the data at a specific level, which is consistent with the idea of deep learning. Mathematically, the standard form of the Taylor series theory can be described below:

Theorem 1: We have an integer $i \geq 0$ and assume a mapping function $f: \mathbb{R} \to \mathbb{R}$ be an i times differentiable form at point $c \in \mathbb{R}$, where $\mathbb{R}$ denotes the real number set. Here, there is a function $g_J: \mathbb{R} \to \mathbb{R}$, so as to satisfy

$$\begin{aligned} f(x) &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 \\ &\quad + \cdots + \frac{f^{(J)}(a)}{J!}(x-a)^J + g_J(x)(x-a)^J \\ &\approx \sum_{i=0}^{J} \frac{f^{(i)}(a)}{i!}(x-a)^i \end{aligned} \tag{1}$$

where $\lim_{x \to a} g_J(x) = 0$; ! is the operator of factorial product.

Considering $f(0) = 0$ and $f^{(i)}(0) = 1$, (1) can be simplified into

$$f(x) = \sum_{i=1}^{J} \frac{1}{i!} x^i. \tag{2}$$

Thus, the matrix form of (2) is given as follows:

$$f(\mathcal{A}) = \sum_{i=1}^{J} \frac{1}{i!} [\mathcal{A}]^i \tag{3}$$

where $[A]^i$ denotes the product of i matrices, which can refer to (7). In the following, we will apply the idea of the Taylor theory to the construction of the deep encoder.

III. Methodology

The developed deep encoder with EKTS for HSI classification will be introduced in detail in this section. Some notes are here given below: $\mathbf{A} \in \mathbb{R}^{M \times N \times L}$ is the input HSI. For convenience, HSI is unfolded into a 2-D matrix $A = \{x_1^T, x_2^T, \ldots, x_k^T, \ldots, x_K^T\} \in \mathbb{R}^{K \times L}$, where $K = M \times N$ and $x_k \in \mathbb{R}^{L \times 1}$ is the pixel vector of $\mathbf{A}$.

A. Problem Statement and Model Construction

As discussed before, deep learning has become the most popular and competitive technique in the task of HSI classification. Most of the current deep-learning-based methods are based on neural networks, such as CNN and DBN, which include many hidden layers (usually more than 2 layers) to find the functions defining the mapping from the input (i.e., HSI data) to the output (i.e., features or labels). For such technique, two issues should be remarked here. *First*, the structure of deep networks is often complex, and there are many neurons in each layer, with the consequence that a large number of free parameters to be trained are included. When training samples are insufficient or unavailable, these deep networks for HSI classification may become performance degradation, even invalid. *Second*, it should be noted that current deep learning methods only use features in the deepest layer to classify HSIs. From the perspective of multiscale analysis, they have a single scale as large as the deepest layer. Although features in the deep layer contain more abstract and robust structure information, some valuable characteristics included in shallow layers are not reflected [30], which can be considered as a kind of information loss.

Motivated by the twofold issues remarked above, in this work, we attempt to construct a deep model without those free parameters that need to be trained, which can simultaneously report and combine shallow features and deep features to predict labels. It could be easily found that for the problem of feature representation or extraction, the idea of the Taylor series is consistent with that of deep learning [35], which can obtain deep features without any free parameters. Also, the Taylor series can also be viewed as a multiscale form, which is able to describe HSIs at different feature levels [34]. The low-level and high-level features can be, respectively, considered as shallow and deep features, and both of them are easily integrated to predict labels.

According to such argument, we can build EKTS to approximate neural networks for extracting deep features of HSIs. More specifically, we first build an observation model for encoding A as follows:

$$\arg\min_{U,H} \ \|A - AHU\|_F^2 + \lambda \|H\|_1 \tag{4}$$

where H and U are the encoding and decoding matrices, respectively. Here, let $H = \sum_{i=1}^{J} H_i$, and thus the input A can be decomposed as J components and each component corresponds to the encoding matrix H_i. As a result, (4) is further rewritten as

$$\arg\min_{U,H_i} \ \left\| A - A\sum_{i=1}^{J} H_i U \right\|_F^2 + \sum_{i=1}^{J} \lambda_i \|H_i\|_1 \tag{5}$$

where λ_i denotes the regularization parameter corresponding to H_i. Equation (5) can be viewed as the procedure of hierarchical decomposition of the HSI A, and each decomposition layer corresponds to the transformation matrix H_i. Considering the role of Taylor series, (3) is introduced into (5), and we thus have

$$\begin{aligned} &\arg\min_{U,H_i} \ \left\| A - A\sum_{i=1}^{J} H_i U \right\|_F^2 + \sum_{i=1}^{J} \lambda_i \|H_i\|_1 \\ &\text{s.t. } AH_i = \frac{1}{i!}[A]^i \overline{\mathbf{1}}, \quad 1 \leq i \leq J \end{aligned} \tag{6}$$

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where $[A]^i$ denotes the product of i matrices, which satisfies

$$[A]^i = \begin{cases} \overbrace{AA^TAA^T, \ldots, AA^T}^{i}, & \text{if } i \text{ is even} \\ \overbrace{AA^TAA^T, \ldots, A^TA}^{i}, & \text{if } i \text{ is odd.} \end{cases} \tag{7}$$

In addition, $\overline{\mathbf{1}}$ denotes the coupling matrix with norm equaling to 1; H_i is the ith encoding matrix of Taylor series, which can be seen an encoding matrix. Note that the objective function of (6) is linear, so that the linear characteristics in the given HSI data cannot be well-described, which is inconsistent with most of the existing deep networks that have the nonlinear ability. To guarantee the nonlinearity, we introduce the kernel theory to impose (6). More specifically, (6) is here transformed into the unconstrained form below

$$\begin{aligned} \underset{U,H_i}{\arg\min} \ & \left\| A - A\sum_{i=1}^{J} H_i U \right\|_F^2 + \sum_{i=1}^{J} \lambda_i \|H_i\|_1 \\ & + \beta \sum_{i=1}^{J} \left\| AH_i - \frac{1}{i!}[A]^i \right\|_F^2. \end{aligned} \tag{8}$$

As indicated by the kernel theory [36], [37], [38], if there are nonlinear relationships among samples in a low-dimensional space, a corresponding high-dimensional space exists so that these sample relationships can be expressed linearly. Suppose $\phi(\cdot)$ as a function, which can map the training set A from the L dimensional space to the O dimensional space, i.e., $A = \{x_1^T, x_2^T, \ldots, x_K^T\} \in \mathbb{R}^{K\times L} \rightarrow \phi(A) = \{\phi(x_1)^T, \phi(x_2)^T, \ldots, \phi(x_K)^T\} \in \mathbb{R}^{K\times O}$, where $O \geq L$. Thus, (8) can be rewritten as the following form:

$$\begin{aligned} \underset{W,G_i}{\arg\min} \ & \left\| \phi(A) - \phi(A)\sum_{i=1}^{J} G_i W \right\|_F^2 + \sum_{i=1}^{J} \lambda_i \|G_i\|_1 \\ & + \beta \sum_{i=1}^{J} \left\| \phi(A)G_i - \frac{1}{i!}[\phi(A)]^i \mathbf{1} \right\|_F^2 \end{aligned} \tag{9}$$

where $G_i \in \mathbb{R}^{O\times P}$ and $W \in \mathbb{R}^{P\times O}$ are, respectively, the encoding and decoding matrices for $\phi(A)$, and P is the dimension of the encoded forms; β is the weight; $\mathbf{1}$ represents the coupling matrix, whose norm is 1, and it is satisfied with

$$\mathbf{1} \in \begin{cases} \mathbb{R}^{K\times P}, & \text{if } i \text{ is even} \\ \mathbb{R}^{O\times P}, & \text{if } i \text{ is odd.} \end{cases} \tag{10}$$

This optimization form is here further transformed into

$$\begin{aligned} \underset{W,G_i}{\arg\min} \ & \phi(A)^T\phi(A) - 2\phi(A)^T\phi(A)\sum_{i=1}^{J} G_i W \\ & + \left(\sum_{i=1}^{J} G_i W\right)^T \phi(A)^T\phi(A)\sum_{i=1}^{J} G_i W \\ & + \sum_{i=1}^{J} \lambda_i \|G_i\|_1 + \beta \\ & \times \sum_{i=1}^{J} \left(G_i^T\phi(A)^T\phi(A)G_i - \frac{2}{i!}G_i^T\phi(A)^T[\phi(A)]^i\mathbf{1} \right. \\ & \left. + \frac{1}{i!\times i!}\mathbf{1}^T\left([\phi(A)]^i\right)^T[\phi(A)]^i\mathbf{1} \right). \end{aligned} \tag{11}$$

Here, we introduce an identity matrix $I \in \mathbb{R}^{O\times O}$. Removing constant items, (11) is simplified into

$$\begin{aligned} \underset{W,G_i}{\arg\min} \ & -2I^T\phi(A)^T\phi(A)I\sum_{i=1}^{J} G_i W \\ & + \left(\sum_{i=1}^{J} G_i W\right)^T I^T\phi(A)^T\phi(A)I\sum_{i=1}^{J} G_i W \\ & + \sum_{i=1}^{J} \lambda_i \|G_i\|_1 + \beta\sum_{i=1}^{J} G_i^T I^T\phi(A)^T\phi(A)IG_j \\ & - \beta\sum_{i=1}^{J} \frac{2}{i!}G_i^T I^T\phi(A)^T[\phi(A)]^i\mathbf{1}. \end{aligned} \tag{12}$$

Theorem 2: If a K-dimensional space is given, which is composed of a vector set of $\{\phi(x_1), \phi(x_2), \ldots, \phi(x_K)\} \in \mathbb{R}^{K\times O}$, any one O-dimensional point $\overrightarrow{\mathbf{p}}$ can be represented as the linear combination of $\phi(x_1), \phi(x_2), \ldots, \phi(x_K)$, that is, $\overrightarrow{\mathbf{p}} = c_1\phi(x_1) + c_2\phi(x_2) + \cdots + c_K\phi(x_K) = \phi(A)^T\mathbf{c}$, where $\mathbf{c} = \{c_1, c_2, \ldots, c_K\}^T \in \mathbb{R}^{K\times 1}$.

According to **Theorem 2**, it can be easily known that the introduced identity matrix $I = [\mathbf{1}_1, \ldots, \mathbf{1}_o, \ldots, \mathbf{1}_O]$ can be expressed by $I = \phi(A)^TC$, where $\mathbf{1}_o = \phi(A)^T\mathbf{c}_o$ and $C = [\mathbf{c}_1, \ldots, \mathbf{c}_o, \ldots, \mathbf{c}_O] \in \mathbb{R}^{K\times O}$. It should be noted that in $\mathbf{1}_o$, one "1" element is included and the rest is "0." Thence, (12) can be transformed into the form with variable C below

$$\begin{aligned} \underset{W,G_i,C}{\arg\min} \ & -2C^T\phi(A)\phi(A)^T\phi(A)\phi(A)^TC\sum_{i=1}^{J} G_i W \\ & + \left(\sum_{i=1}^{J} G_i W\right)^T C^T\phi(A)\phi(A)^T\phi(A)\phi(A)^T \\ & \times C\sum_{i=1}^{J} G_i W + \sum_{i=1}^{J} \lambda_i \|G_i\|_1 \\ & + \beta\sum_{i=1}^{J} \left(G_i^T C^T\phi(A)\phi(A)^T\phi(A)\phi(A)^TCG_i\right) \\ & - \beta\sum_{i=1}^{J} \frac{2}{i!}G_i^T C^T\phi(A)\phi(A)^T[\phi(A)]^i\mathbf{1}. \end{aligned} \tag{13}$$

Considering the inner product of $\phi(A)\phi(A)^T$, the kernel function $\mathcal{K}$ is naturally introduced, where $\mathcal{K} = \phi(A)\phi(A)^T \in \mathbb{R}^{K\times K}$. However, since the component $[\phi(A)]^i$ is included in (13), the value of i is fixed to be even so that the kernelwise operation is successfully performed. So, (13) is rewritten as

$$\begin{aligned} \underset{W,G_j,C}{\arg\min} \ & -2C^T\mathcal{K}^2C\sum_{j=1}^{J/2} G_j W \\ & + \left(\sum_{j=1}^{J/2} G_j W\right)^T C^T\mathcal{K}^2C\sum_{j=1}^{J/2} G_j W \\ & + \sum_{j=1}^{J/2} \lambda_j \|G_j\|_1 + \beta\sum_{j=1}^{J/2} \left(G_j^T C^T\mathcal{K}^2CG_j\right) \\ & - \beta\sum_{j=1}^{J/2} \left(\frac{2}{(2j)!}G_j^T C^T\mathcal{K}^{j+1}\mathbf{1}\right) \end{aligned} \tag{14}$$

where $j = i/2$ ($i \in \{2, 4, \ldots, J\}$, $j \in \{1, 2, \ldots, J/2\}$) for $\lambda_j = \lambda_{i/2}$ and $G_j = G_{i/2}$. This model is called the deep

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encoder with EKTS, aiming to extract deep features of HSIs by estimating G_j.

B. Optimization Strategy

As can be known that (14) is convex. Here, we design an alternating iterative optimization strategy to estimate the solutions W, G_j, and C of (14). To be specific, we here introduce a set of auxiliary variables $\mathcal{V}_j = G_j$ ($j = \{1, 2, 3, \ldots, J/2\}$, where J is even). Accordingly, (14) approximates the following form:

$$\begin{aligned}\underset{W,G_j,C,\mathcal{V}_j}{\arg\min} \quad & -2C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_jW \\ & +\left(\sum_{j=1}^{J/2}G_jW\right)^T C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_jW \\ & +\beta\sum_{j=1}^{J/2}\left(G_j^TC^T\mathcal{K}^2CG_j-\frac{2}{(2j)!}G_j^TC^T\mathcal{K}^{j+1}\mathbf{1}\right) \\ & +\sum_{j=1}^{J/2}\lambda_j\|\mathcal{V}_j\|_1+\gamma\sum_{j=1}^{J/2}\|\mathcal{V}_j-G_j\|_F^2 \end{aligned} \tag{15}$$

where γ is the coupling parameter. For (15), we optimize each variable by fixing others. Thus, W, G_j, C, and $\mathcal{V}_j$ are, respectively, optimized as follows:

1) Optimizing W: Fixing G_j, C, and $\mathcal{V}_j$, W can be optimized by

$$\begin{aligned}\underset{W}{\arg\min} \quad & -2C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_jW \\ & +\left(\sum_{j=1}^{J/2}G_jW\right)^T C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_jW. \end{aligned} \tag{16}$$

Here, we use the gradient decent strategy to obtain the solution of (16). Concretely, we take the derivative of the objective function in (16) with respect to W, that is,

$$\begin{aligned}\frac{\partial\mathcal{L}(W)}{\partial W} = & -2\left(C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_j\right)^T \\ & +2\left(\sum_{j=1}^{J/2}G_j\right)^T C^T(\mathcal{K}^2)^TC\sum_{j=1}^{J/2}G_jW \end{aligned} \tag{17}$$

where $\mathcal{L}(W)$ denotes the objective function in (16).

Theorem 3: For a given kernel matrix $\mathcal{K} = \phi(A)\phi(A)^T$, since the inner product operation is symmetrical, the kernel matrix $\mathcal{K}$ is also symmetrical. Furthermore, the square of kernel matrix $\mathcal{K}^2$ is still symmetrical.

According to **Theorem 3**, (17) is equivalent to

$$\begin{aligned}\frac{\partial\mathcal{L}(W)}{\partial W} = & -2\left(C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_j\right)^T \\ & +2\left(\sum_{j=1}^{J/2}G_j\right)^T C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_jW. \end{aligned} \tag{18}$$

Let $((\partial\mathcal{L}(W))/\partial W) = 0$, and W for this subproblem can thus be calculated via

$$W=\left(\left(\sum_{j=1}^{J/2}G_j\right)^T C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_j\right)^{-1}\left(C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_j\right)^T. \tag{19}$$

2) Optimizing G_j: Fix W, C, and $\mathcal{V}_j$, and G_j is computed by the following subproblem:

$$\begin{aligned}\underset{G_j}{\arg\min} \quad & -2C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_jW \\ & +\left(\sum_{j=1}^{J/2}G_jW\right)^T C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_jW \\ & +\beta\sum_{j=1}^{J/2}\left(G_j^TC^T\mathcal{K}^2CG_j-\frac{2}{(2j)!}G_j^TC^T\mathcal{K}^{j+1}\mathbf{1}\right) \\ & +\gamma\sum_{j=1}^{J/2}\|\mathcal{V}_j-G_j\|_F^2. \end{aligned} \tag{20}$$

We take the derivative of the objective function in (20) with respect to G_j, that is,

$$\begin{aligned}\left.\frac{\partial\mathcal{L}(G_j)}{\partial G_j}\right|_{j=r} = & -2\left(C^T\mathcal{K}^2C\right)^TW^T \\ & +2\left(C^T\mathcal{K}^2CG_rWW^T+C^T\mathcal{K}^2C\sum_{j\neq r}G_jWW^T\right) \\ & +2\beta\left(C^T\mathcal{K}^2CG_r-\frac{1}{(2r)!}C^T\mathcal{K}^{r+1}\mathbf{1}\right) \\ & +2\gamma\left(G_r-\mathcal{V}_r\right) \end{aligned} \tag{21}$$

where $\mathcal{L}(G_j)$ is the objective function in (21). Let $((\partial\mathcal{L}(G_j))/\partial G_j)|_{j=r} = 0$, and G_j can be computed by

$$\begin{aligned}& C^T\mathcal{K}^2CG_rWW^T+\left(\beta C^T\mathcal{K}^2C+\gamma I\right)G_r \\ & \quad =\left(C^T\mathcal{K}^2C\right)^TW^T-C^T\mathcal{K}^2C\sum_{j\neq r}G_jWW^T \\ & \qquad +\frac{1}{(2r)!}\beta C^T\mathcal{K}^{r+1}\mathbf{1}+\gamma\mathcal{V}_r. \end{aligned} \tag{22}$$

Equation (22) is the Sylvester equation, which can be solved by conjugate gradient (CG) [39], [40]. Here, three points should be remarked for CG. *First*, the system matrix associated with (22) is symmetric and positive definite, and it provides the necessary condition for the application of CG. *Second*, the heaviest computation for CG lies in the multiplication of the system matrix times a vector, which can be efficiently implemented in the matrix representation. *Third*, CG is capable of achieving the convergence with only a few iterations. Actually, such convergence is linear and depends on the ratio between the largest and the smallest eigenvalues of the system matrix associated with (22) [41]. In our experimental study, about 30 iterations can report a very good approximation of the solution of (22).

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3) Optimizing C: When W, G_j, and $\mathcal{V}_j$ are fixed, we can obtain the subproblem with regarding to C as following:

$$\begin{aligned}\arg\min_{C} \quad & -2C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_jW \\ & +\left(\sum_{j=1}^{J/2}G_jW\right)^T C^T\mathcal{K}^2C\sum_{j=1}^{J/2}G_jW \\ & +\beta\sum_{j=1}^{J/2}\left(G_j^TC^T\mathcal{K}^2CG_j-\frac{2}{(2j)!}G_j^TC^T\mathcal{K}^{j+1}\mathbf{1}\right).\end{aligned} \tag{23}$$

The corresponding derivative is provided as

$$\begin{aligned}\frac{\partial\mathcal{L}(C)}{\partial C}=&-2\left(\mathcal{K}^2C\sum_{j=1}^{J/2}G_jW+\mathcal{K}^2CW^T\left(\sum_{j=1}^{J/2}G_j\right)^T\right) \\ &+2\beta\sum_{j=1}^{J/2}\left(\mathcal{K}^2CG_jG_j^T-\frac{1}{(2j)!}\mathcal{K}^{j+1}\mathbf{1}G_j^T\right)\end{aligned} \tag{24}$$

where $\mathcal{L}(C)$ is the objective function in (23). Let $((\partial\mathcal{L}(C))/\partial C)=0$, and we have

$$\begin{aligned}\mathcal{K}^2C\left(\sum_{j=1}^{J/2}G_jW+W^T\left(\sum_{j=1}^{J/2}G_j\right)^T+\beta\sum_{j=1}^{J/2}G_jG_j^T\right) \\ =\beta\sum_{j=1}^{J/2}\frac{1}{(2j)!}\mathcal{K}^{j+1}\mathbf{1}G_j^T.\end{aligned} \tag{25}$$

Equation (25) is also satisfied with the Sylvester form, which can be directly solved via CG.

4) Optimizing $\mathcal{V}_j$: Keeping W, G_j, and C fixed, the subproblem with variable $\mathcal{V}_j$ is found below

$$\arg\min_{\mathcal{V}_j}\ \sum_{j=1}^{J/2}\lambda_j\|\mathcal{V}_j\|_1+\gamma\sum_{j=1}^{J/2}\|\mathcal{V}_j-G_j\|_F^2 \tag{26}$$

which yields the closed-form solution

$$\mathcal{V}_j=\text{soft}\left(G_j,\frac{\lambda_j}{2\gamma}\right) \tag{27}$$

where $\text{soft}(a,b)=\text{sign}(a)*\max(|a|-b,0)$.

By the fourfold optimization operations, the final W, G_j, C, and $\mathcal{V}_j$ can be obtained once the convergence condition is satisfied. Thus, we use G_j and C to get the discriminative features Z, that is,

$$Z=\left[\mathcal{K}CG_1,\mathcal{K}CG_2,\ldots,\mathcal{K}CG_j,\ldots\right] \tag{28}$$

where $Z\in\mathbb{R}^{K\times S/2}$ ($S=J/2\times P$). As can be seen from (28), the final feature form Z is obtained by stacking outputs $\mathcal{K}CG_j$ from all the layers. In other words, Z combines shallow features and deep features, which contains richer information than the features obtained by the current deep learning strategies with only the deepest layer.

C. SVM Classifier for Label Prediction

We here recall HSI data $A\in\mathbb{R}^{K\times L}$. The corresponding feature form is $Z\in\mathbb{R}^{K\times S/2}$, which can be obtained by (28). To predict the labels of HSI A, the SVM is used to classify the corresponding feature Z. More specifically, first, the training set Z_a with K_a samples is selected from Z, and the rest is the testing set Z_e with K_b samples. Second, the training set Z_a assigned with labels is used to train the SVM model, whose purpose is to seek the optimal separating hyperplane maximizing the margin between the closest features. The boundary points, which are called the support vectors, are used for constructing a decision surface. Third, the trained SVM model is used to predict any unlabeled samples. Any unlabeled features from Z_e can be assigned with a label by the relative distance to the hyperplane, and the decision function is described into

$$\text{Label}(z)=\text{sign}\left(\sum_{i=1}^{K_a}s_i\mathcal{A}_i\overline{\mathcal{K}}((Z_a)_i,z)+h\right) \tag{29}$$

where $\text{Label}(z)$ is the predicted label of z; h denotes the decision offset coefficient determined via the trained s, following the Karush-Kuhn-Tucker (KKT) condition [42]; $\overline{\mathcal{K}}$ is a kernel function. Here, to classify multiclass targets in HSI, the one-versus-rest (OVR) strategy [43] is used to integrate multiple binary SVM classifiers.

IV. Experiment

A. Experimental Datasets

To assess the performance of our EKTS method, we perform experiments on the Salinas, Indian Pines, and Pavia Centre datasets, which can be found at http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes. Salinas was originally imaged via the 224-band airborne/visible infrared imaging spectrometer (AVIRIS) sensor on Salinas Valley, USA, which has the spatial resolution of 512×217. We reduce the amount of bands to 204 after removing the 20 water absorption bands. For the sake of decreasing computational consumption, the Salinas data are downsampled by a 2 factor to acquire the 256×108 spatial resolution. There exist 16 classes in this ground truth (see Fig. 2). The Indian Pines dataset was also collected by AVIRIS, whose target scene is located at Indian Pines from northwest Indiana. Similarly, the 20 water absorption bands have been removed so as to retain the 204 bands with 145×145 spatial resolution. The corresponding ground truth contains 16 classes (see Fig. 3). The Pavia Centre was recorded via the reflective optics system imaging spectrometer (ROSIS) sensor at the location of the center of Pavia. After removing noise bands and downsampling all the bands, the updated data with $365\times238\times103$ were obtained, where the ground truth has nine classes (see Fig. 4).

B. Quantitative Metrics

We use three popular metrics to quantitatively evaluate the classification performance, including overall accuracy (OA), average accuracy (AA), and Kappa coefficient. The OA index

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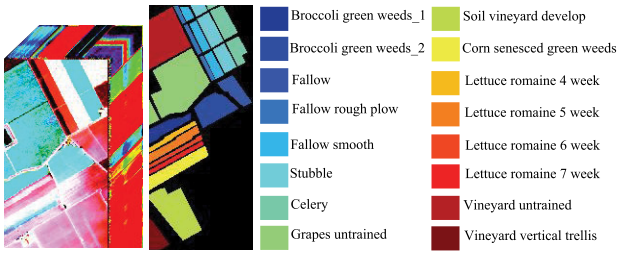


Fig. 2. Salinas data and their ground-truth map.



Fig. 3. Indian Pines data and their ground-truth map.

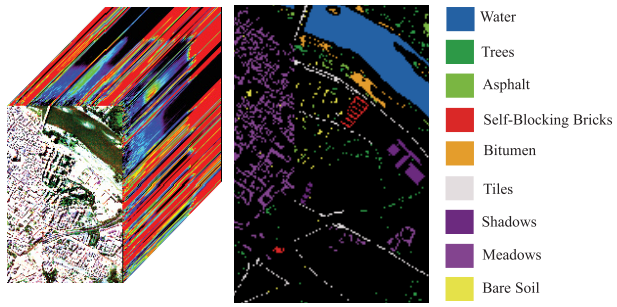


Fig. 4. Pavia Centre data and their ground-truth map.

is the percentage of pixels identified correctly. The AA metric is used to reflect the average percentage of pixels for each class. The Kappa coefficient is also adopted for reflecting the classification accuracy, which is based on the confusion matrix.

C. Parameter Analysis

To explore the performance of the proposed EKTS approach, it is necessary to conduct the parameter analysis. $\lambda_j (1 \leq j \leq J/2)$ is used to control the sparsity of the encoding matrix in (14). β is the weight of the Taylor constraint. J is even, and $J/2$ denotes the depth of the constructed network. In addition, the kernel $\mathcal{K}$ in our method is set to be Gaussian in our experiments, since the Gaussian kernel is nonlinear and strictly positive definite, which has only one predetermined parameter, that is, the kernel width. Therefore, we also explore the influence of the kernel width on the performance of the proposed approach. To successfully perform the parameter analysis, the Salinas data are used as a testing example, and the OA, AA, and Kappa metrics are used to measure the classification results under different parameter values, which are shown in Figs. 5 and 6.

1) Analysis of Parameter λ_j: For convenient parameter analysis, we set all λ_j to the same value λ, i.e., $\lambda_j = \lambda$ $(1 \leq j \leq J/2)$. Thus, the analysis of λ_j can be simplified into that of λ, with the results shown in Fig. 5(a), where λ is selected from the range of $[0.0001, 0.001, 0.01 0.1, 1, 10]$. It is found that the OA, AA, and Kappa metrics achieve the largest values at $\lambda = 0.1$. When $\lambda > 0.1$ and $\lambda < 0.1$, the OA, AA, and Kappa are decreasing. The reason is summarized below: A relative small value of λ (i.e., $\lambda < 0.1$) means the weakened sparsity of the encoding matrix, which is no doubt to cause the performance degradation of our proposed EKTS method. On the contrary, a too big value of λ (i.e., $\lambda > 0.1$) will increase the risk of information loss in feature extraction.

2) Analysis of Parameter β: Fig. 5(b) presents the influence of parameter β on the proposed EKTS method (in OA, AA, and Kappa), where β is also selected from $[0.0001, 0.001, 0.01 0.1, 1, 10]$. From this figure, we can see that the EKTS method achieves the best classification performance (in OA, AA, and Kappa) at $\beta = 0.1$, as well as obviously degraded performance at $\beta > 0.1$ and $\beta < 0.1$. On one hand, if a small value of β is used, the contribution of the Taylor constraint becomes low, which reduces the ability to perceive information of the input HSI. On the other hand, if a too large value of β is used, the weight of such constraint becomes overlarge, which reduces the importance of other components in (9).

3) Analysis of Parameter J: In Fig. 5(c), the influence of parameter J on the EKTS method (in OA, AA, and Kappa) is visually presented, where J is selected from $[2, 4, 6, 8, 10, 12]$. As shown in this figure, the EKTS method with $J = 6$ lies in the biggest OA, AA, and Kappa values. A small value of J, such as $J = 2$ and $J = 4$, means that relatively few network layers are used, which have no ability to extract deep features of the input HSI. In contrast, if too many layers are included, that is, $J > 6$, a small disturbance like noise in HSI may have a significant impact on the classification results; meanwhile, the calculation consumption will also be increased.

4) Analysis of Kernel Width Parameter: Fig. 6 is adopted to present the influence of the kernel width parameter of the Gaussian kernel $\mathcal{K}$ on the performance (in OA, AA, and Kappa) of our EKTS method, where the width values are selected from $[2, 4, 8, 16, 32, 64, 128]$. As can be observed, the best performance is achieved when the width is fixed to 8. When the width value is smaller than 8, the classification performance tends to be decreasing. The reason is that the smaller the width value, the more sensitive the kernel is to the input HSI data, which may largely increase the negative influence of small anomalies (such as noise) contained in the data on the classification performance. When the width value is bigger than 8, the performance of the proposed EKTS method tends to become degenerative since the data perception ability of the kernel is decreasing.

It should be noted that these results of parametric analysis on the Salinas dataset are also applicable to the Indian Pines and Pavia Centre datasets. λ, β, J, and kernel width of our EKTS method are fixed to 0.1, 0.1, 6, and 8, respectively in

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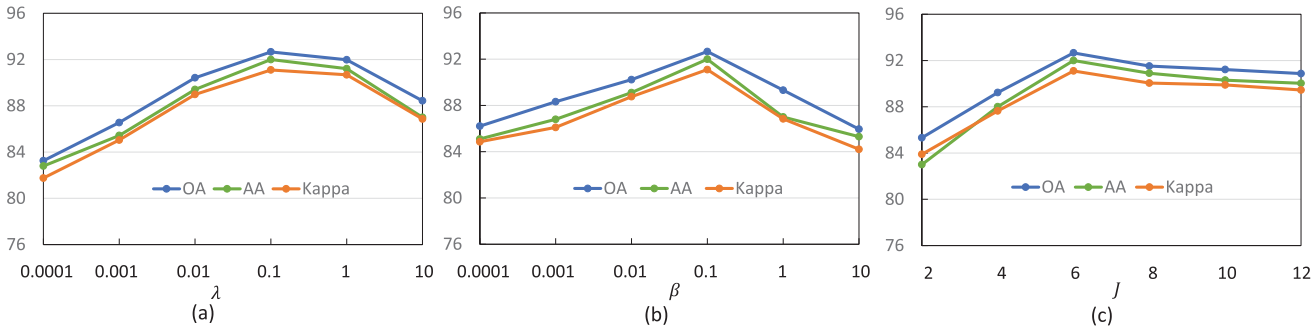


Fig. 5. Parameter influence of (a) λ, (b) β, and (c) J on the performance (in OA, AA, and Kappa) of our EKTS method.

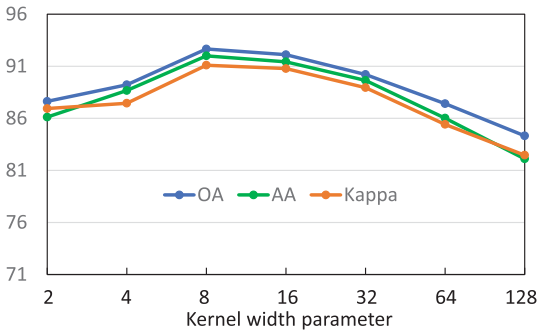


Fig. 6. Influence of the kernel width parameter of Gaussian kernel $\mathcal{K}$ on the performance (in OA, AA, and Kappa) of our EKTS method.

all the follow-up experiments including algorithm comparison and ablation studies.

D. Algorithm Comparison

1) Competing Approaches: In our experimental study, the algorithm comparison is indispensable to validate the superiority of the proposed method. The used competing methods include RAW-SVM [5], EMAP [12], random patches network (RPNet) [30], CNN-AL [21], SMBN [26], CDSAE [17], ENL-FCN [27], and RSEN [28]. For the RAW-SVM method, the original spectral features of HSIs are directly used for the SVM classifier to predict labels. The EMAP is mathematical morphology-based, which constructs morphological profiles with the first three principal components to extract features. RPNet, CNN-AL, and ENL-FCN are CNN-based. The SMBN uses a squeeze multibias network to incorporate multibias into the rectified linear unit (ReLU) layers. The CDSAE is based on the autoencoder (AE) network. The RSEN is a recent popular self-ensembling network-based. On the whole, both RAW-SVM and EMAP belong to the traditional machine learning methods, and RPNet, CNN-AL, SMBN, CDSAE, ENL-FCN, and RSEN are recent deep-learning-based.

2) Parameter Settings: For fair comparison, the parameter selections of all the competing methods are first suggested by the original authors, and we further empirically fine-tune these methods to reach the best performance. In addition, in our EKTS method, the parameters λ_j, β, J, and kernel width are, respectively, set to 0.1, 0.1, 6, and 8, following the results of parameter analysis in Section IV-C.

3) Training Sample Illustration: To achieve comprehensive comparison on the Salinas, Indian Pines, and Pavia Centre datasets, the ratio of training samples is successively fixed to 1%, 5%, 10%, 15%, and 20% from each dataset, and the rest is used as the testing samples. It should be noted that for our EKTS-based classification method, it is not an end-to-end learning method, which includes EKTS for feature extraction and SVM for label prediction. Actually, EKTS for feature extraction has no any free parameters that need to be trained and does not require any training samples. Therefore, the above selected training samples are not prepared for EKTS in feature extraction, but only for SVM in label prediction and those competing approaches.

4) Results and Analysis: We repeat the experiment 10 times for each method and report the averaged values of OA, AA, and Kappa in Tables I–III. In these three tables, the best OA, AA, and Kappa results are marked in **bold**. As can be found from these tables, all the metric values of the proposed EKTS method are bigger compared with those of RAW-SVM, EMAP, RPNet, CNN-AL, SMBN, CDSAE, ENL-FCN, and RSEN, except for the AA value of CDSAE with 20% ratio of the training samples on the Indian Pines dataset. In other words, the classification results of the proposed EKTS method are basically better than the results of other competing approaches in different training samples' cases.

In addition, we visually present the classification maps to more comprehensively compare the performance of different methods, with the results shown in Figs. 7–9, where 10% ratio of the training samples from each dataset is used. From these three figures, it is seen that the classification maps of our EKTS approach, which have less misidentified pixels, are more closely related to the ground truths than those of RAW-SVM, EMAP, RPNet, CNN-AL, SMBN, CDSAE, ENL-FCN, and RSEN. For instance, in Fig. 7, the "Grapes untrained" and "Vineyard untrained" classes of our EKTS [see Fig. 7(j)] are obviously clearer, which look like in a more smooth scene with less "errors," compared with those of other methods [see Fig. 7(b)–(i)]. In addition, it can be easily found from the green closeups in Figs. 8 and 9 that our EKTS method obtains more consistent results with the ground truths than RAW-SVM, EMAP, RPNet, CNN-AL, SMBN, CDSAE, ENL-FCN, and RSEN.

According to the above-presented experimental results and related analysis, it can be concluded that our EKTS method

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TABLE I

METRIC RESULTS OF DIFFERENT METHODS ON THE SALINAS DATASET

Ratio of Training Samples	Metrics	Classification Methods								
		RAW-SVM	EMAP	RPNet	CNN-AL	SMBN	CDSAE	ENL-FCN	RSEN	EKTS
1%	OA(%)	64.49	71.31	88.24	89.55	88.53	90.21	89.32	87.01	**93.24**
	AA(%)	70.79	79.65	84.94	90.36	88.12	91.21	87.39	87.66	**94.55**
	Kappa×100	57.89	66.83	86.54	88.01	87.54	89.65	89.01	87.24	**92.87**
5%	OA(%)	87.61	93.93	95.66	96.92	97.04	98.01	96.95	96.88	**98.23**
	AA(%)	90.11	96.24	97.72	97.79	98.04	98.71	98.01	97.75	**98.85**
	Kappa×100	86.10	93.20	95.18	96.57	96.70	97.78	96.51	96.63	**98.03**
10%	OA(%)	92.70	94.60	96.64	97.65	97.99	97.98	97.38	96.61	**99.00**
	AA(%)	96.05	97.59	97.95	98.30	98.76	98.83	98.42	98.06	**99.24**
	Kappa×100	91.81	93.88	96.26	97.39	97.76	97.75	97.08	95.66	**98.88**
15%	OA(%)	92.88	96.65	97.66	97.74	97.68	97.97	98.17	98.75	**99.31**
	AA(%)	96.19	98.11	98.66	98.25	98.61	98.69	98.97	99.05	**99.37**
	Kappa×100	92.00	96.25	97.40	97.48	97.42	97.74	98.44	98.98	**99.23**
20%	OA(%)	92.89	97.36	98.11	98.43	98.15	98.54	98.65	98.01	**99.35**
	AA(%)	96.46	98.05	98.74	98.96	98.76	99.22	99.32	98.82	**99.57**
	Kappa×100	92.01	97.04	97.90	98.25	97.94	98.38	98.49	98.33	**99.28**

TABLE II

METRIC RESULTS OF DIFFERENT METHODS ON THE INDIAN PINES DATASET

Ratio of Training Samples	Metrics	Classification Methods								
		RAW-SVM	EMAP	RPNet	CNN-AL	SMBN	CDSAE	ENL-FCN	RSEN	EKTS
1%	OA(%)	70.01	78.55	85.43	87.23	86.43	87.94	87.67	88.62	**90.30**
	AA(%)	48.07	56.58	66.42	66.68	67.94	70.34	69.43	70.72	**76.78**
	Kappa×100	65.07	75.05	83.29	85.31	84.41	86.17	85.85	87.06	**88.87**
5%	OA(%)	74.81	84.74	91.38	93.58	93.54	94.51	93.92	92.77	**95.67**
	AA(%)	64.09	66.12	77.33	83.63	82.29	87.83	86.04	82.06	**88.55**
	Kappa×100	71.14	82.56	90.12	92.65	92.61	93.71	93.05	91.73	**95.05**
10%	OA(%)	82.13	93.38	95.75	95.50	96.21	96.84	96.26	96.37	**96.88**
	AA(%)	70.39	86.81	92.29	90.82	90.70	93.34	93.62	93.43	**94.25**
	Kappa×100	79.48	92.45	95.15	94.87	96.17	96.39	95.73	95.90	**96.44**
15%	OA(%)	85.30	94.37	96.61	96.07	96.49	97.33	97.04	97.25	**97.46**
	AA(%)	76.33	89.73	92.31	92.91	91.37	94.49	93.67	94.57	**95.04**
	Kappa×100	83.18	93.56	96.13	95.51	95.99	96.95	96.62	96.86	**97.10**
20%	OA(%)	86.71	94.85	97.17	97.56	98.26	98.09	97.99	97.98	**98.18**
	AA(%)	83.26	90.32	93.08	94.28	94.16	**97.00**	95.18	95.29	94.97
	Kappa×100	84.81	94.11	96.77	97.22	98.01	97.82	97.71	97.69	**97.93**

TABLE III

METRIC RESULTS OF DIFFERENT METHODS ON THE PAVIA CENTRE DATASET

Ratio of Training Samples	Metrics	Classification Methods								
		RAW-SVM	EMAP	RPNet	CNN-AL	SMBN	CDSAE	ENL-FCN	RSEN	EKTS
1%	OA(%)	78.81	81.33	90.35	90.05	91.33	90.60	92.11	91.19	**93.08**
	AA(%)	32.61	68.32	77.45	79.92	83.21	82.14	85.15	84.89	**85.66**
	Kappa×100	67.98	71.51	86.30	85.54	87.12	86.41	88.73	87.34	**90.14**
5%	OA(%)	84.56	89.19	94.35	93.72	95.40	95.68	95.14	95.65	**96.44**
	AA(%)	74.36	81.33	87.47	83.32	86.11	86.35	86.18	84.28	**87.17**
	Kappa×100	76.43	84.19	91.98	91.09	93.50	93.88	93.12	93.82	**94.95**
10%	OA(%)	86.80	90.22	95.22	97.78	96.07	97.84	98.48	97.96	**98.52**
	AA(%)	76.85	82.80	92.01	92.68	93.14	93.39	94.39	93.20	**95.90**
	Kappa×100	80.33	86.13	93.27	96.86	94.45	96.94	97.85	97.11	**97.91**
15%	OA(%)	87.83	92.84	96.83	98.33	98.41	98.57	98.44	98.04	**98.72**
	AA(%)	77.98	86.56	92.73	94.84	94.83	94.86	95.75	94.62	**96.04**
	Kappa×100	81.37	89.40	95.51	97.63	97.75	97.97	97.79	97.23	**98.19**
20%	OA(%)	88.05	94.19	97.30	98.65	98.77	98.81	98.39	98.17	**98.92**
	AA(%)	78.69	88.12	93.13	95.90	96.24	95.60	94.54	95.14	**97.08**
	Kappa×100	81.78	91.52	96.18	98.09	98.26	98.31	97.72	97.42	**98.48**

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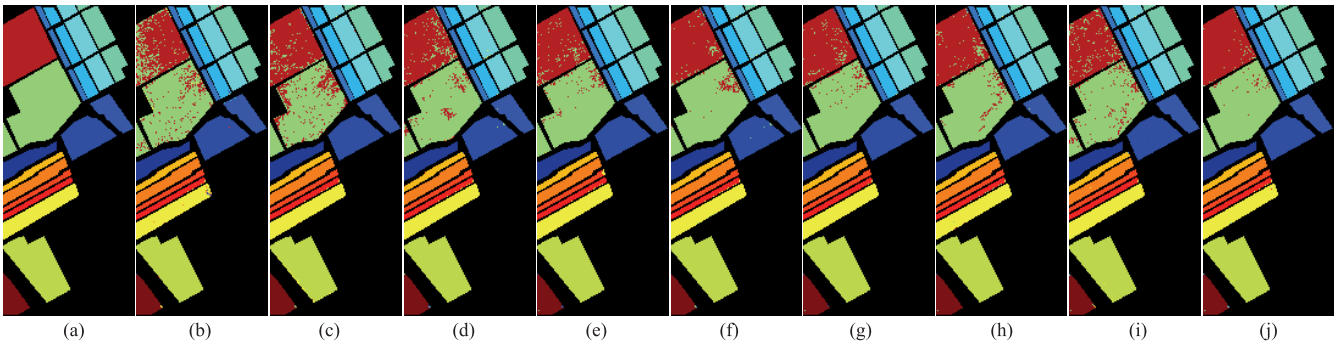


Fig. 7. Classification maps of different methods with 10% ratio of training samples on the Salinas data. (a) Ground truth. (b) RAW-SVM. (c) EMAP. (d) RPNet. (e) CNN-AL. (f) SMBN. (g) CDSAE. (h) ENL-FCN. (i) RSEN. (j) Proposed EKTS.

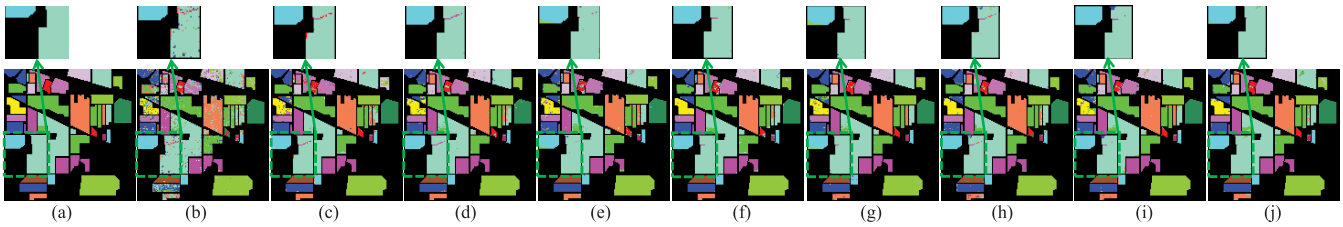


Fig. 8. Classification maps of different methods with 10% ratio of training samples on the Indian Pines data. (a) Ground truth. (b) RAW-SVM. (c) EMAP. (d) RPNet. (e) CNN-AL. (f) SMBN. (g) CDSAE. (h) ENL-FCN. (i) RSEN. (j) Proposed EKTS.

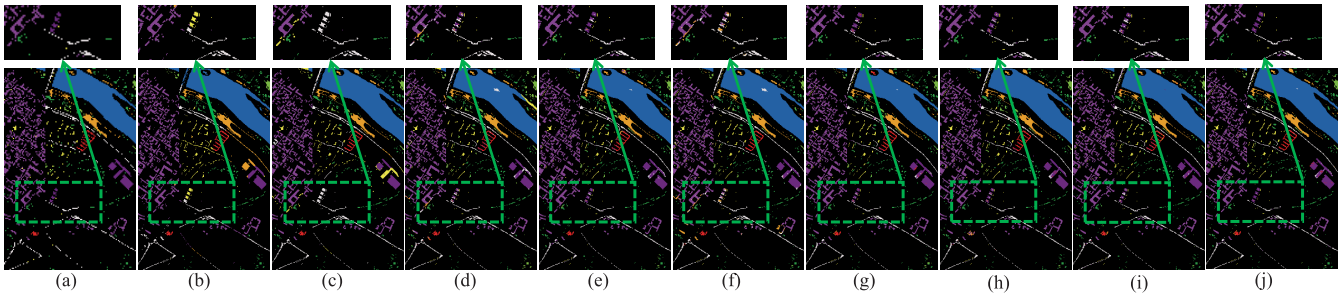


Fig. 9. Classification maps of different methods with 10% ratio of training samples on the Pavia Centre data. (a) Ground truth. (b) RAW-SVM. (c) EMAP. (d) RPNet. (e) CNN-AL. (f) SMBN. (g) CDSAE. (h) ENL-FCN. (i) RSEN. (j) Proposed EKTS.

outperforms RAW-SVM, EMAP, RPNet, CNN-AL, SMBN, CDSAE, ENL-FCN, and RSEN. The reasons can be summarized as follows. RAW-SVM and EMAP are based on shallow features, which are usually not competitive with those deep-learning-based methods. Though RPNet, CNN-AL, SMBN, CDSAE, ENL-FCN, and RSEN are based on deep learning, which have the strong ability to extract and use the deep features hidden for classification, they neglect the role of shallow features. Unlike these approaches, our EKTS method not only effectively extracts deep features of HSIs but also integrates both shallow and deep features for the classification task. In addition, in our experiments, EKTS in feature extraction is implemented without any training samples, which is an important advantage over current deep-learning-based methods.

Here, to further investigate the performance of the proposed EKTS method, we compare the computational efficiency of different approaches, with the results shown in Table IV, where each presented value is the averaged run time of each approach over the Salinas, Indian Pines, and Pavia Centre datasets. As can be seen from this table, the averaged run time of RAW-SVM, EMAP, and RPNet is less than that of our EKTS method. The main reason is that these compared approaches have relatively simple structure, however, which cannot guarantee to report high-quality discriminative features. Furthermore, we can also find that the AL-CNN, SMBN, CDSAE, ENL-FCN, and RSEN approaches spend far more time than our EKTS, since they need so much time and massive resources to train networks, which have high computational consumption. Considering the balance between classification accuracy and computational efficiency, it is finally concluded that the proposed EKTS method achieves more satisfactory classification performance compared with these state-of-the-art approaches.

E. Discussion

In this section, we will conduct fourfold ablation studies to discuss the effectiveness of the Taylor series, the kernel,

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TABLE IV
AVERAGED TIME OF EACH METHOD OVER THE SALINAS, INDIAN PINES, AND PAVIA CENTRE DATASETS

Methods	RAW-SVM	EMAP	RPNet	CNN-AL	SMBN	CDSAE	ENL-FCN	RSEN	EKTS
Time Cost (/s)	1.5 s	6.1 s	19.5 s	490.9 s	687.4 s	1509.9 s	454.7 s	251.2 s	57.3 s

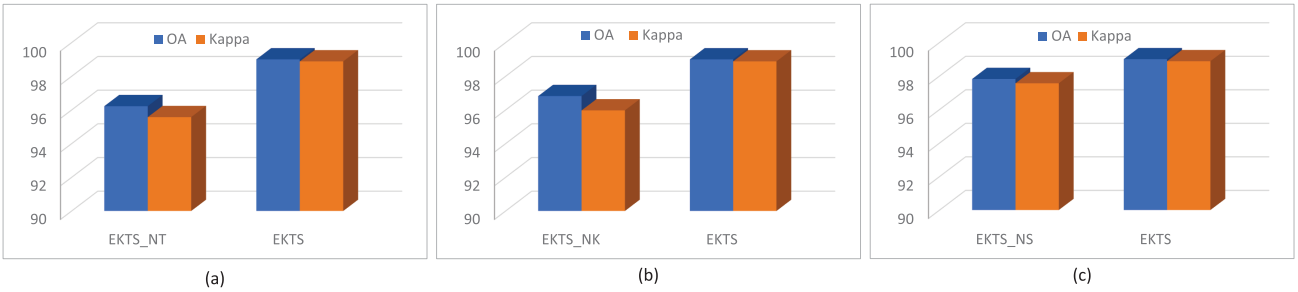


Fig. 10. EKTS is compared with (a) EKTS form excluding the Taylor series (denoted by EKTS_NT); (b) EKTS form excluding kernel (denoted by EKTS_NK); and (c) EKTS form excluding the combination of shallow and deep features (denoted by EKTS_NS).

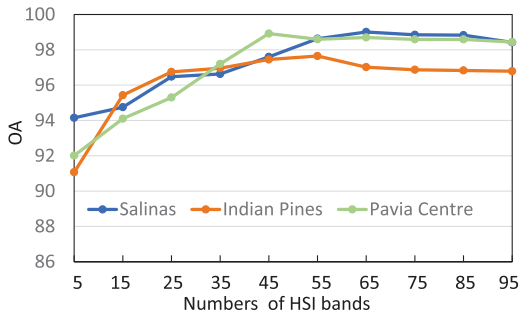


Fig. 11. Impact of the number of HSI bands on the classification results (in OA) of our EKTS method.

and the combination of shallow and deep features in our EKTS, as well as the impact of the number of HSI bands on the classification results of our method. For convenient presentation, the effectiveness of the Taylor series, the kernel, and the combination of shallow and deep feature is explored on the Salinas dataset, with the results shown in Fig. 10. The impact of the number of HSI bands on the classification results of our method is investigated on the Salinas, Indian Pines, and Pavia Centre datasets, which is visually shown in Fig. 11. About 10% training samples are used for SVM included in our EKTS classification method.

1) Effectiveness of Taylor Series: In our EKTS method, the Taylor series is used to approximate neural networks so that deep features of the HSI data can be obtained. We compare EKTS with the EKTS form excluding the Taylor series (denoted by EKTS_NT) to demonstrate the effectiveness of such Taylor series, with the results presented in Fig. 10(a). From this subfigure, we directly observe that the OA and Kappa values of EKTS_NT are clearly smaller than those of EKTS, which indicates that the Taylor series can increase the classification performance of EKTS. Namely, the Taylor series is effective and valuable in our EKTS method.

2) Effectiveness of Kernel: The kernel in our built deep encoder is used to mine and extract nonlinear features of the HSI data. To validate the effectiveness of the kernel, our EKTS approach is compared with the EKTS form excluding the kernel (simplified as EKTS_NK), where the results are shown in Fig. 10(b). It can be found that the EKTS method for HSI classification acquires the bigger OA and Kappa values compared with EKTS_NK. Thus, the kernel included in our EKTS is effective to extract nonlinear features for the promotion of the classification performance.

3) Effectiveness of Combination of Shallow Features and Deep Features: We combine both shallow features and deep features to determine the final features for label prediction. To explore the effectiveness of such combination, we compare the proposed EKTS with the form only including the features at the deepest layer (denoted by EKTS_NS), with the results provided in Fig. 10(c). As can be directly found that the OA and Kappa results of EKTS_NS are worse compared with those of EKTS, which implies the performance improvement achieved by the combination of shallow features and deep features. In other words, combining shallow features and deep features has obvious effectiveness for EKTS in the HSI classification task.

4) Impact of Number of HSI Bands on Results of Our EKTS Method: It is well-known that the number of HSI bands carries the capacity of spectral information, which has impact on the classification results. To quantitatively explore such impact, we observe the classification results (in OA) of our EKTS method under different numbers of HSI bands in Fig. 11, where the number range is $[5, 15, 25, 35, 45, 55, 65, 75, 85, 95]$. From this figure, it can be found that our EKTS method achieves the best classification results (in OA) at specific numbers of HSI bands, which are 65, 55, and 45 on Salinas, Indian Pines, and Pavia Centre, respectively. When the numbers of HSI bands are smaller than these specific values, the classification results become bad, which is caused by insufficient spectral information. On the contrary, when the numbers of HSI bands are larger than the specific values, the classification results also tend to be bad due to the increasing amount of spectral redundant information. Therefore, the number of HSI bands has a significant effect on the classification results of our EKTS method.

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V. Conclusion

In this article, we develop a novel deep encoder with EKTS for the classification of HSIs. Concretely, the Taylor series is introduced to approximate the role of neural network as a way to extract deep features, for the sake of avoiding free parameters that need to be trained. Furthermore, the kernel theory is used to obtain EKTS to mine and extract deep nonlinear features of the HSI data. We construct an alternating iterative optimization strategy to optimize the deep encoder model with EKTS, so that the outputs of all the layers are obtained. Subsequently, the outputs of all the layers are stacked, so as to integrate both shallow features and deep features together as the final features. These final features are finally added in the SVM classifier for predicting the labels of the input HSI. Different from the existing deep learning methods, our EKTS has no free parameters to be trained and combines the advantages of both shallow and deep features to predict labels. We evaluate the performance of the developed EKTS method on three public HSI datasets, with results verifying its effectiveness and superiority in HSI classification.

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